

Population aging and intergenerational relationships

Teachers

Elisenda Rentería (Eli) - erenteria@ced.uab.es

Júlia Almeida Calazans - jalmeida@ced.uab.es

Mariana Calderón-Jaramillo – mcalderon@ced.uab.es



Program

1. **Introduction to Population Aging:** Definition, Trends, and Measures, Demographic Dividend
2. **Economic implications:** Economic profiles, National Transfers Accounts, Social Programs, Alternative views
3. **Health implications:** Life expectancy and Healthy Life expectancy, Healthcare utilization, Active aging
4. **Social implications:**



Every day

1. Lecture
2. Break – 10m (around 11.15h)
3. Lecture
4. Practical example or discussion – 20/30m



What is Aging?



Population Aging

A demographic shift where the proportion of older adults (typically those 65 and over) in a population increases, while the proportion of younger people declines

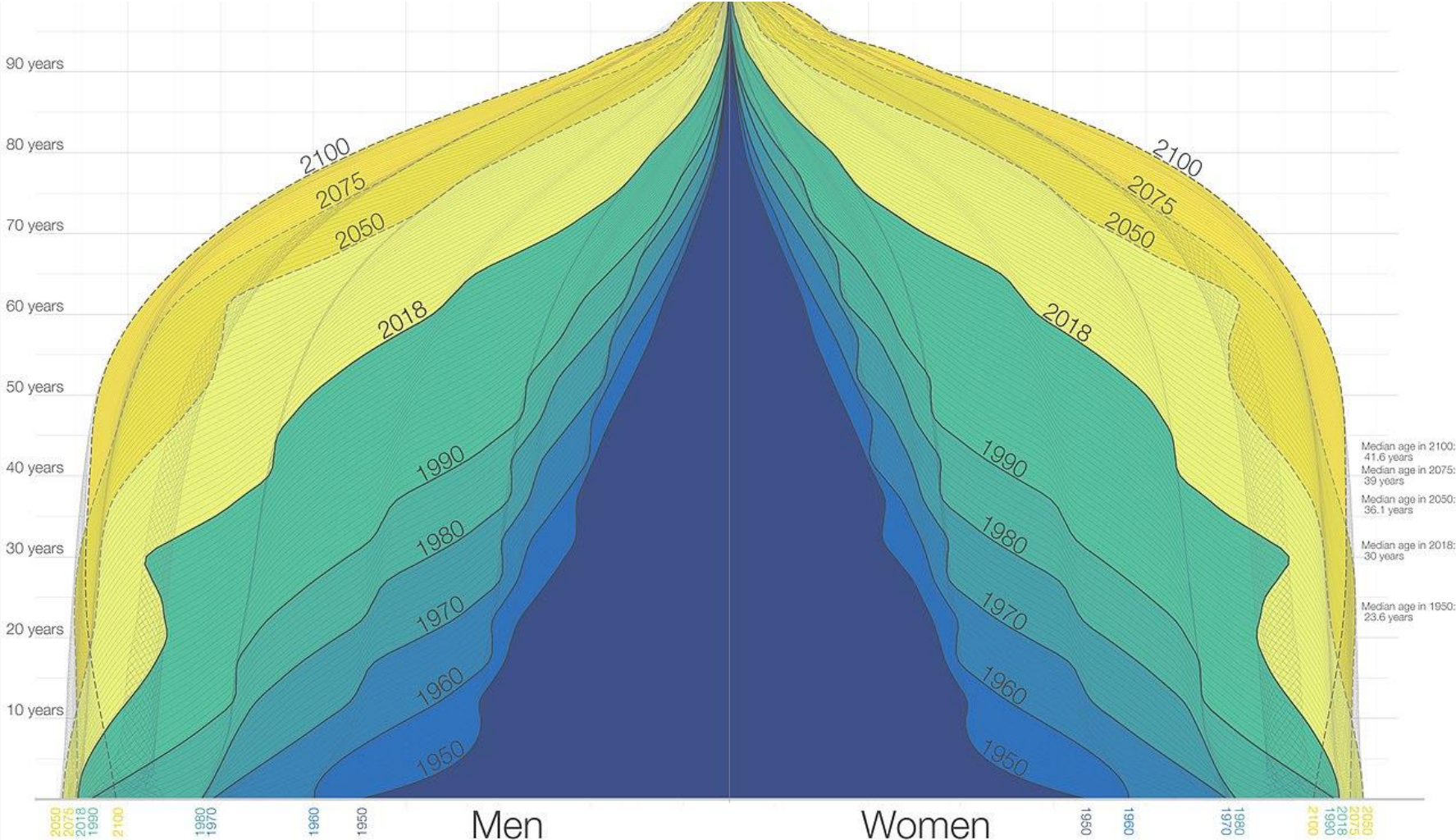
≠

Individual Aging

The biological, psychological, and social changes that occur as people age.

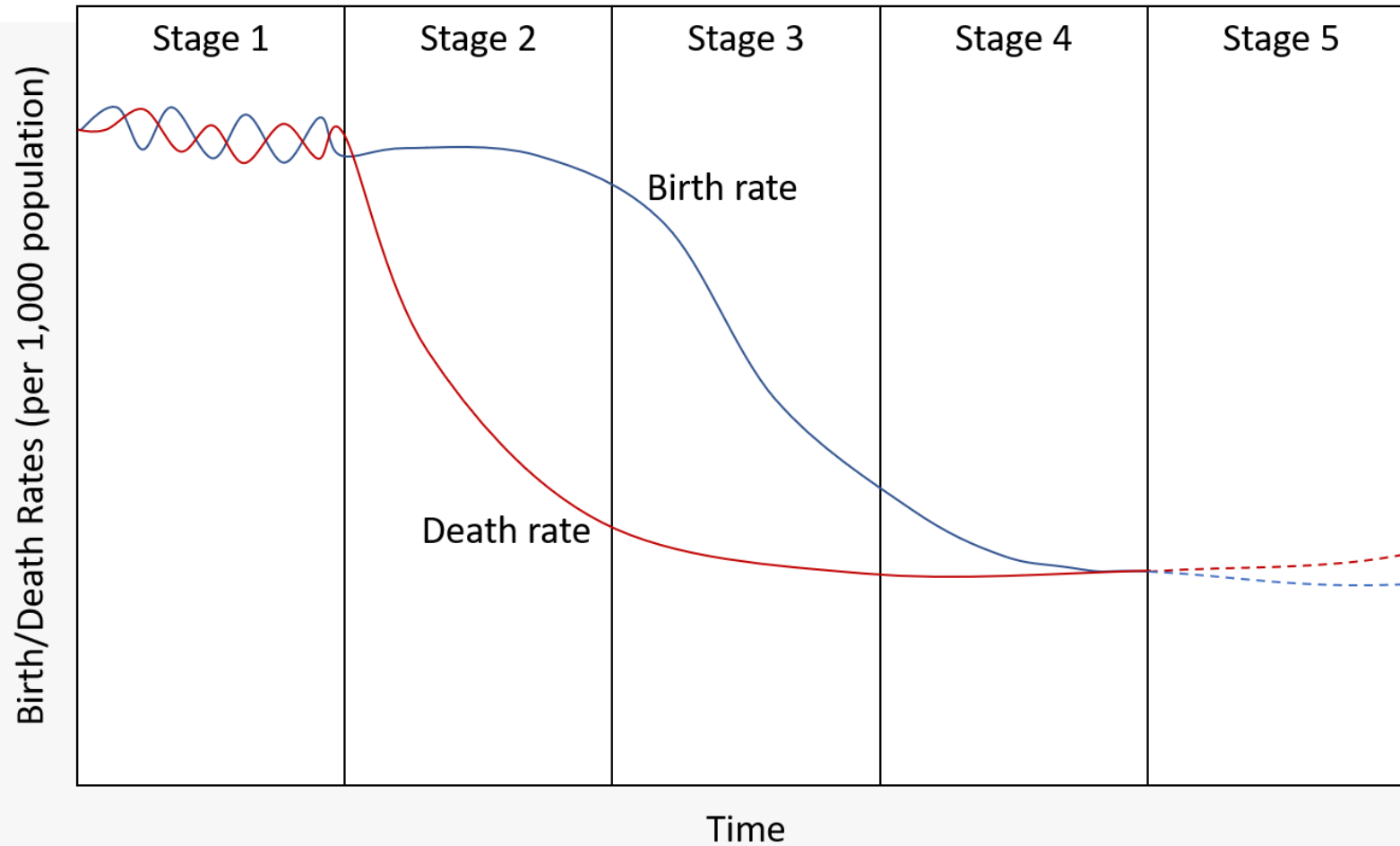


The world is aging



Data source: United Nations Population Division – World Population Prospects 2017; Medium Variant.
The data visualization is available at [OurWorldinData.org](https://ourworldindata.org), where you find more research on how the world is changing and why.
Licensed under CC-BY by the author Max Roser.

Components of Population Aging – The demographic transition



Components of Population Aging

- Mortality / Fertility / Migration
- Fertility has been pointed out as the main component, but mortality is becoming more important (Scott and Canudas-Romo, 2024)
- Other researchers say mortality is the main component (De Santis and Salinari, 2023).
- Open debate, but all components interrelate with each other, and it depends on their levels and which ages they are affecting.



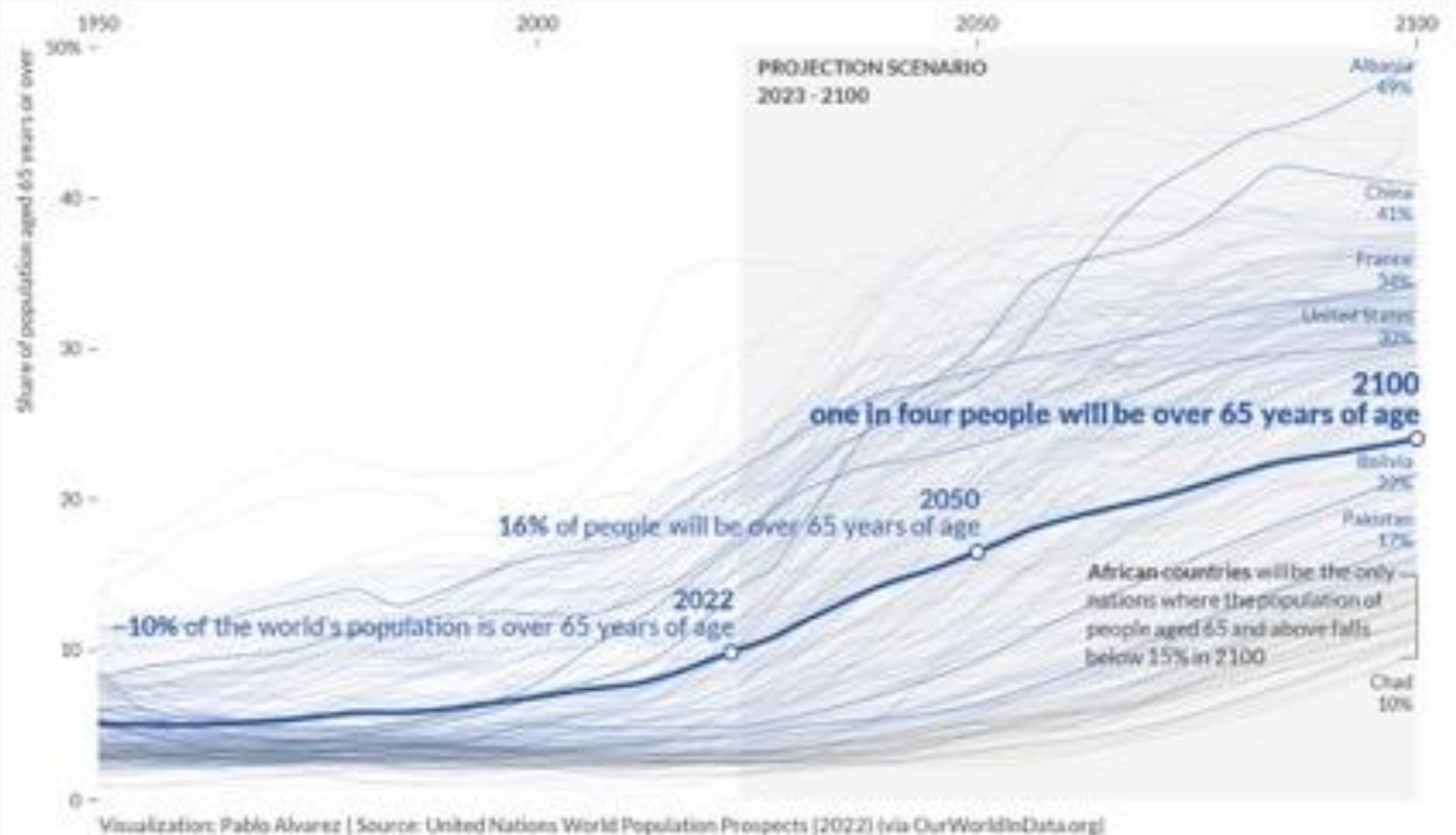
How a country ages?



Age pyramid of Spain



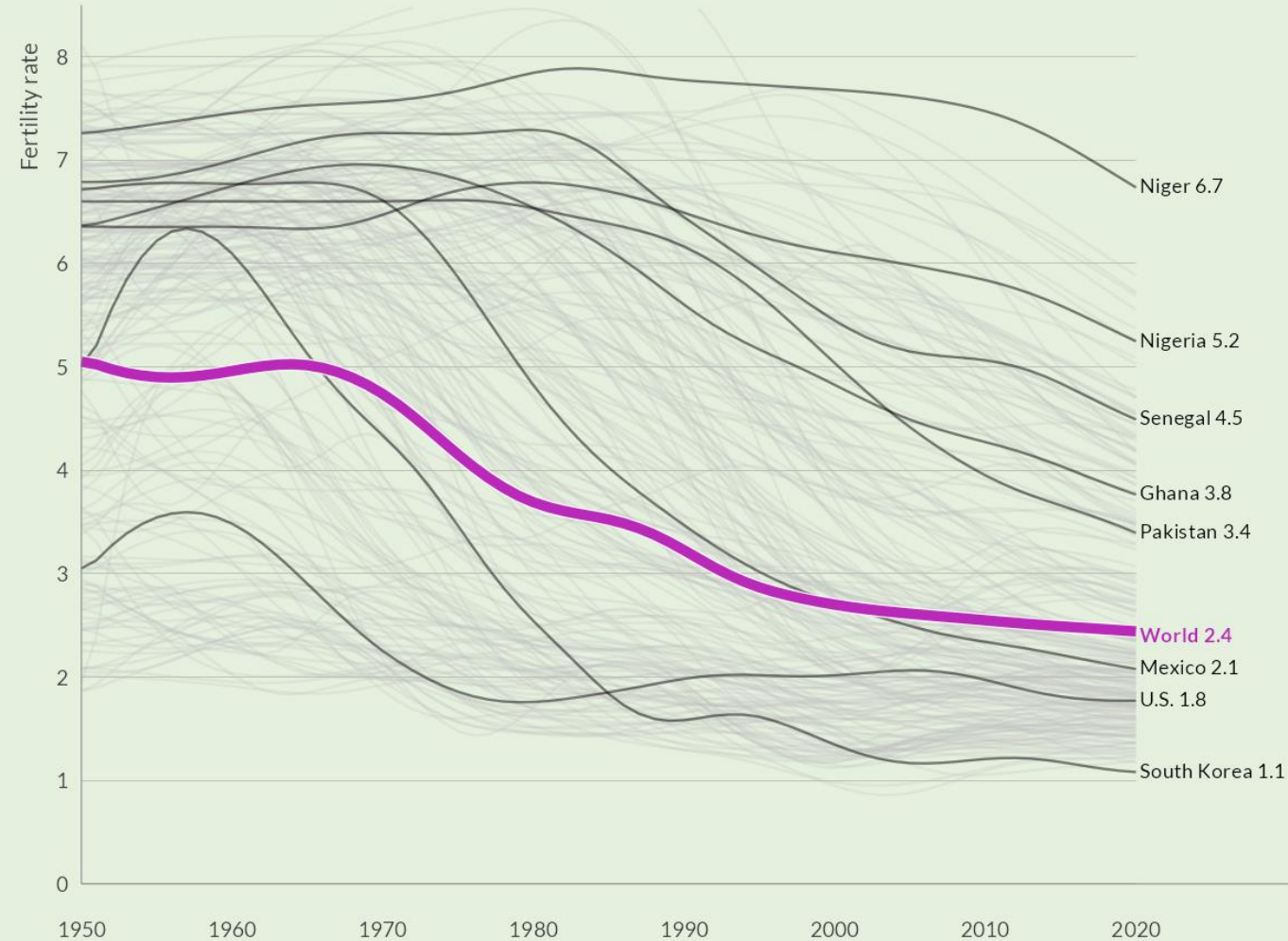
Diverse aging



Different speeds in fertility declines

The Global Decline of the Fertility Rate

70 years ago the average woman had five children, since then the number has halved

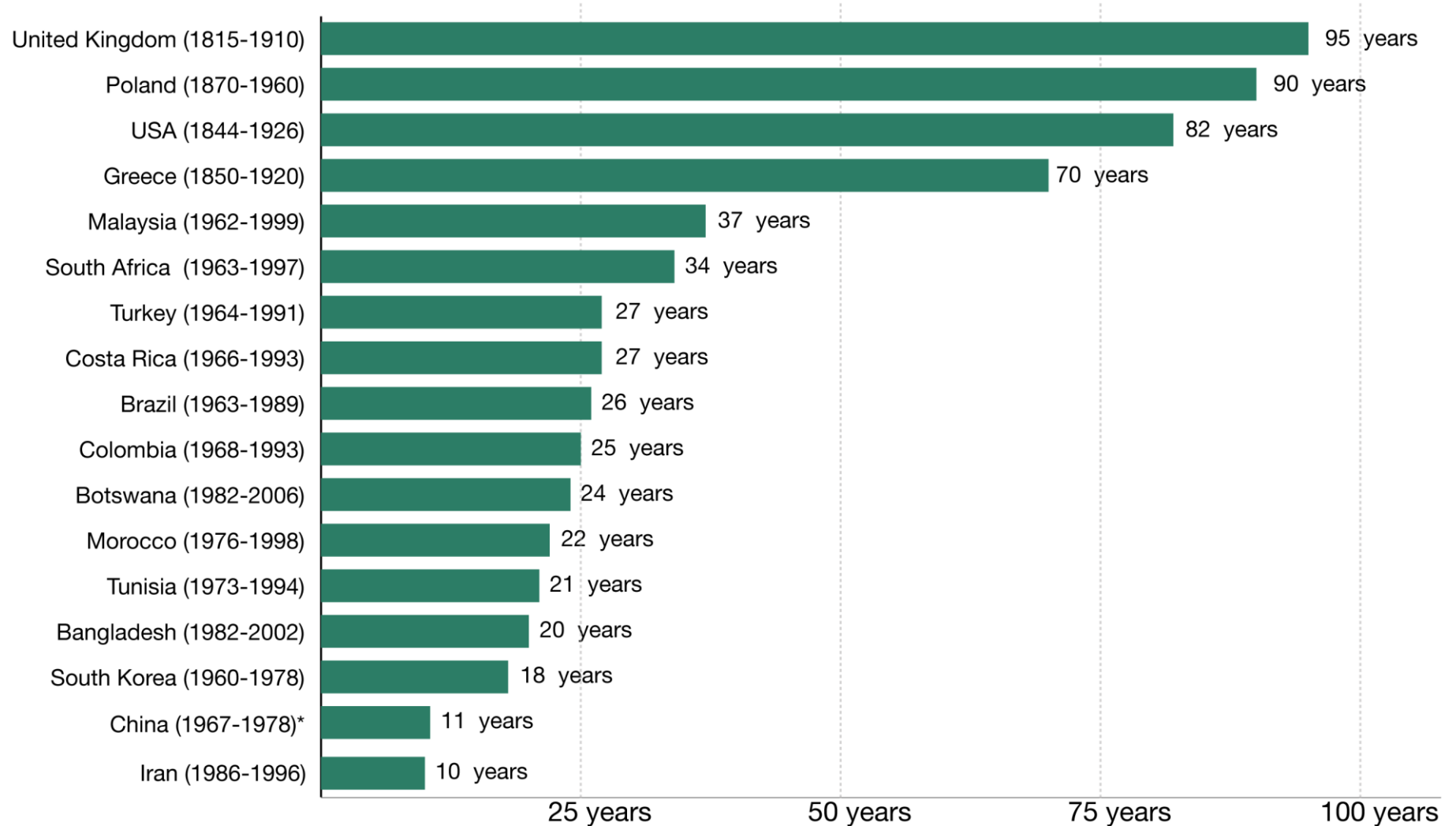


Visualization by Pablo Alvarez | Data from Our World In Data
Fertility rate measures the average number of children per woman



How long did it take for fertility to fall from more than 6 children per woman to fewer than 3 children per woman?

Different speeds in fertility declines



* The one-child-policy in China was introduced after the decline of the total fertility rate below 3. It was introduced between 1978 and 1980.

Data source: The data on the total fertility rate is taken from the Gapminder fertility dataset (version 6) and the World Bank World Development Indicators.

The interactive data visualization is available at [OurWorldinData.org](https://ourworldindata.org). There you find the raw data and more visualizations on this topic.

Licensed under [CC-BY-SA](https://creativecommons.org/licenses/by-sa/4.0/) by the author Max Roser.



Generational equilibrium



Why it is important to measure population aging?

1. Challenges

- Healthcare systems
- Pension schemes
- Long-term care services



Why it is important to measure population aging?

2. Benefits

- Healthier populations
- Intergenerational solidarity
- Social integration



What is important to measure to monitor population aging?

1. Labor activity rates
2. Health status
3. Public expenditures
4. Gender inequalities



How to measure population aging



Dependency Ratios

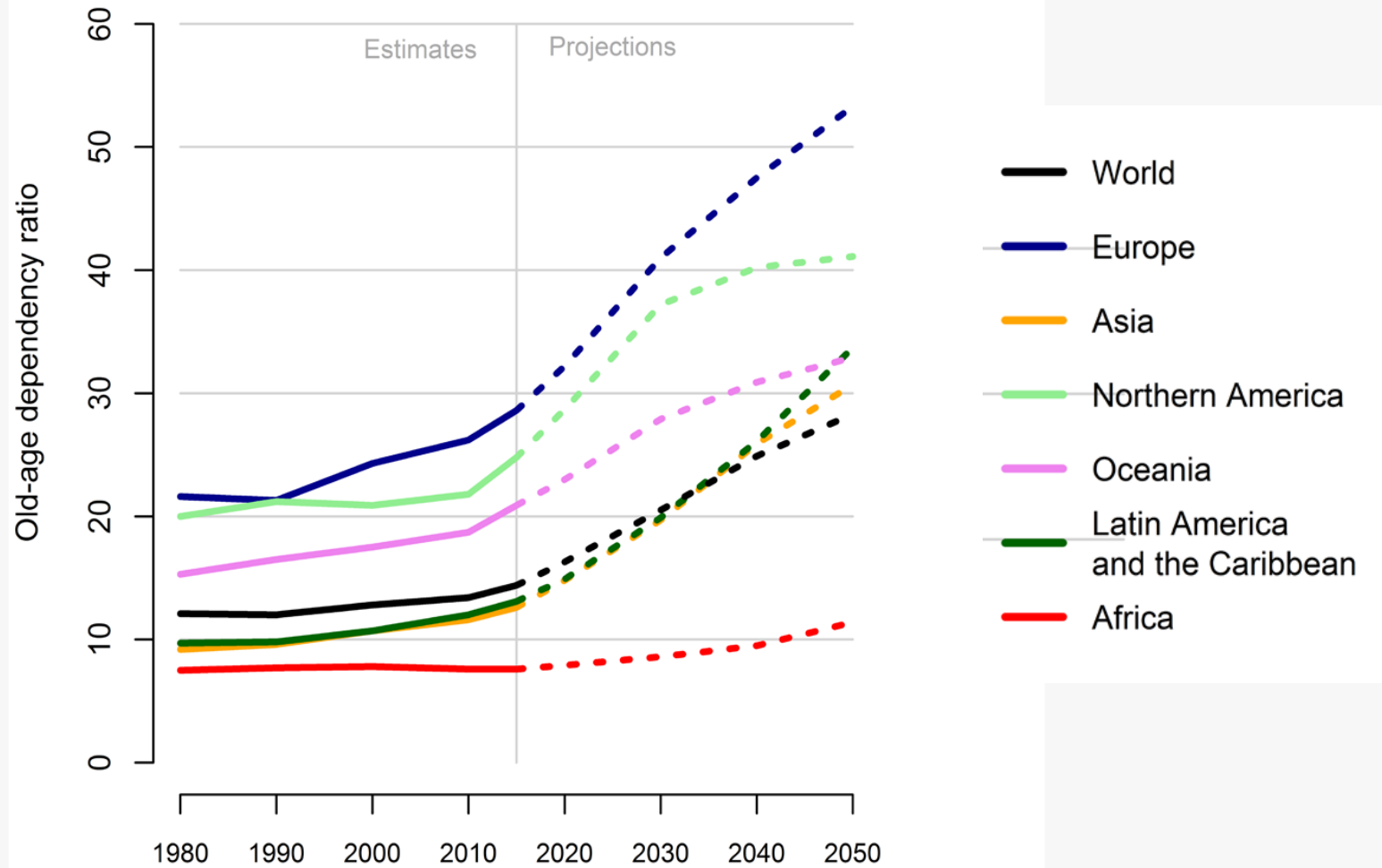
Total Dependency Ratio = Old age dependency ratio + Youth dependency ratio

$$\text{Old age dependency ratio (OADR)} = \frac{\text{Population 65 years old and +}}{\text{Population 15 – 64 years old}} \times 100$$

$$\text{Youth dependency ratio (YDR)} = \frac{\text{Population 0 – 14 years old}}{\text{Population 15 – 64 years old}} \times 100$$



Dependency Ratios



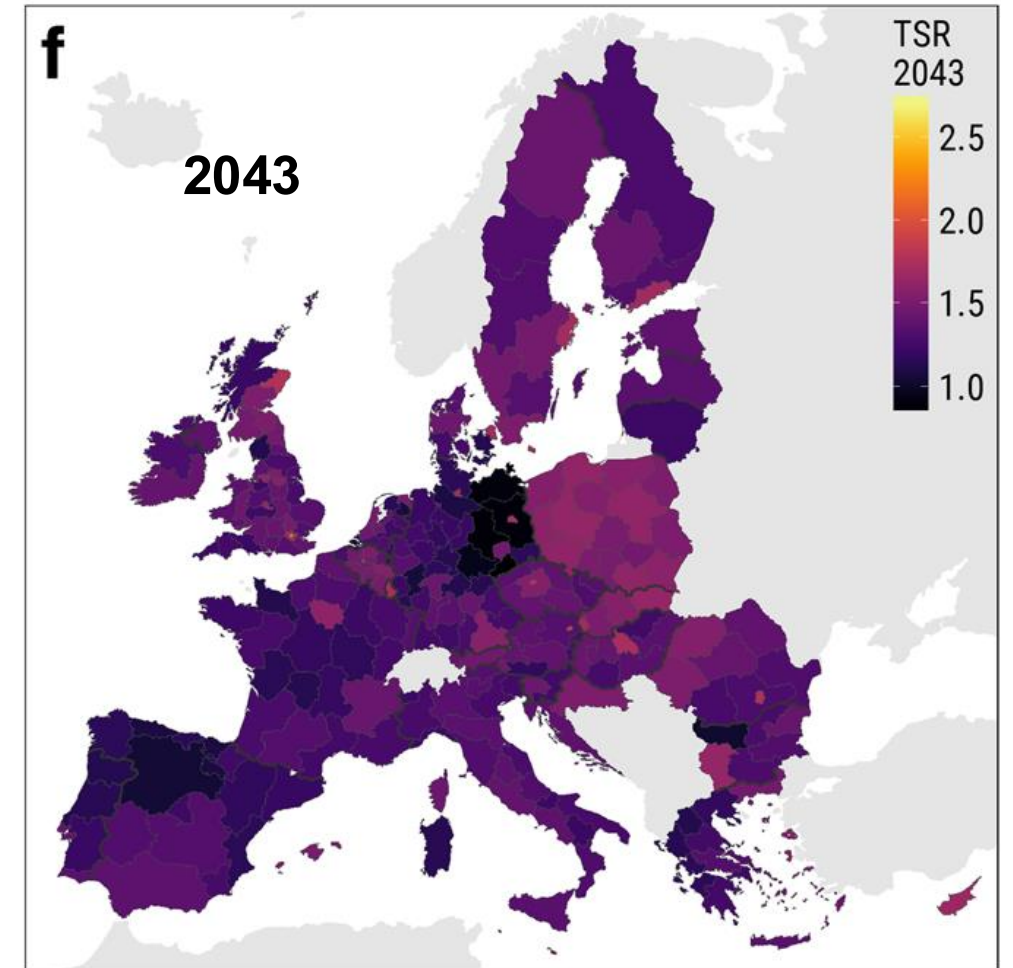
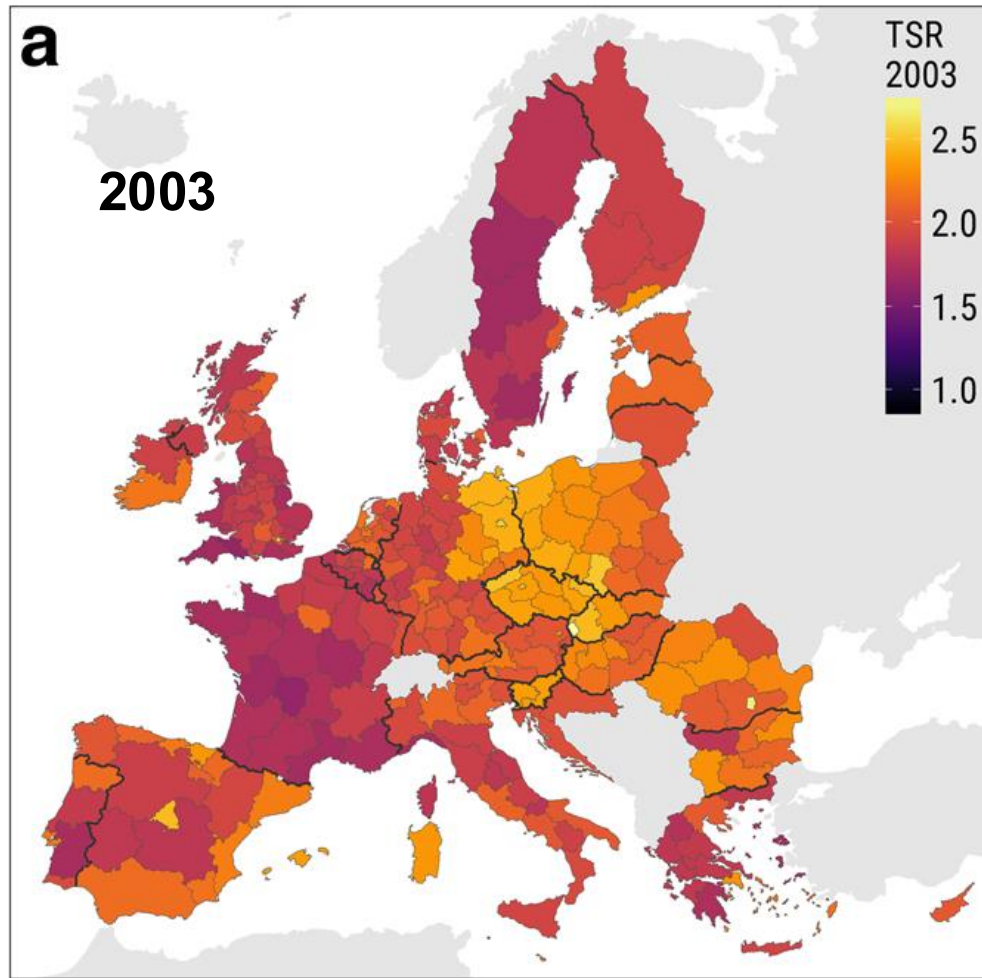
United Nations. World Population Aging, 2017

Support Ratio

$$\text{Support Ratio (SR)} = \frac{\text{Population 15 – 64 years old}}{\text{Population 0 – 14} + \text{Population 65} +} \times 100$$

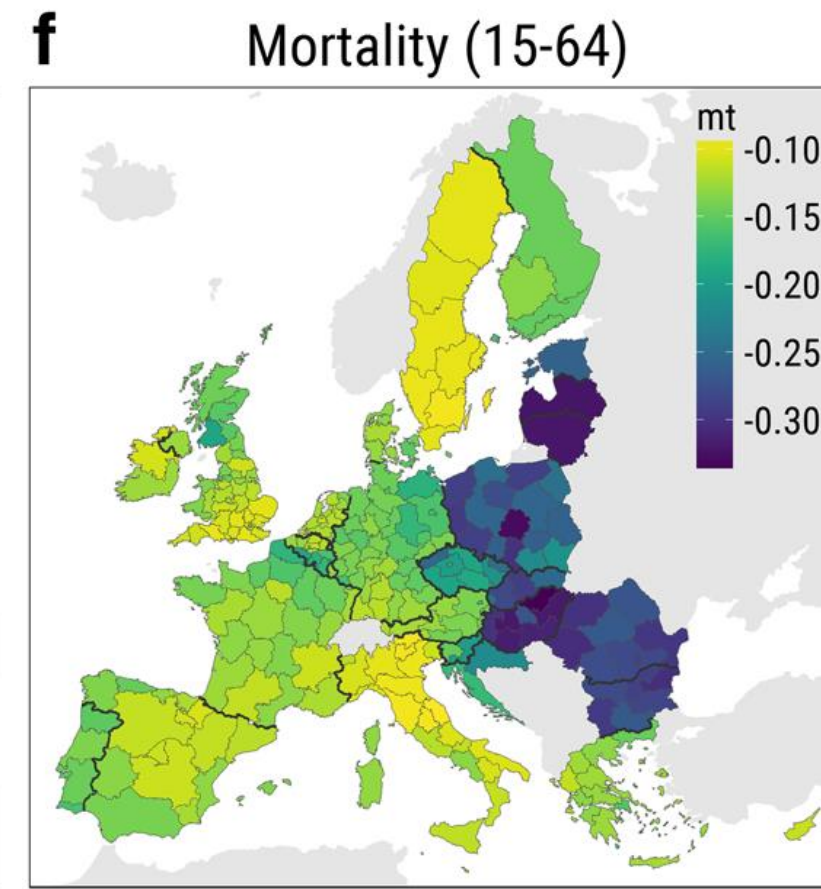
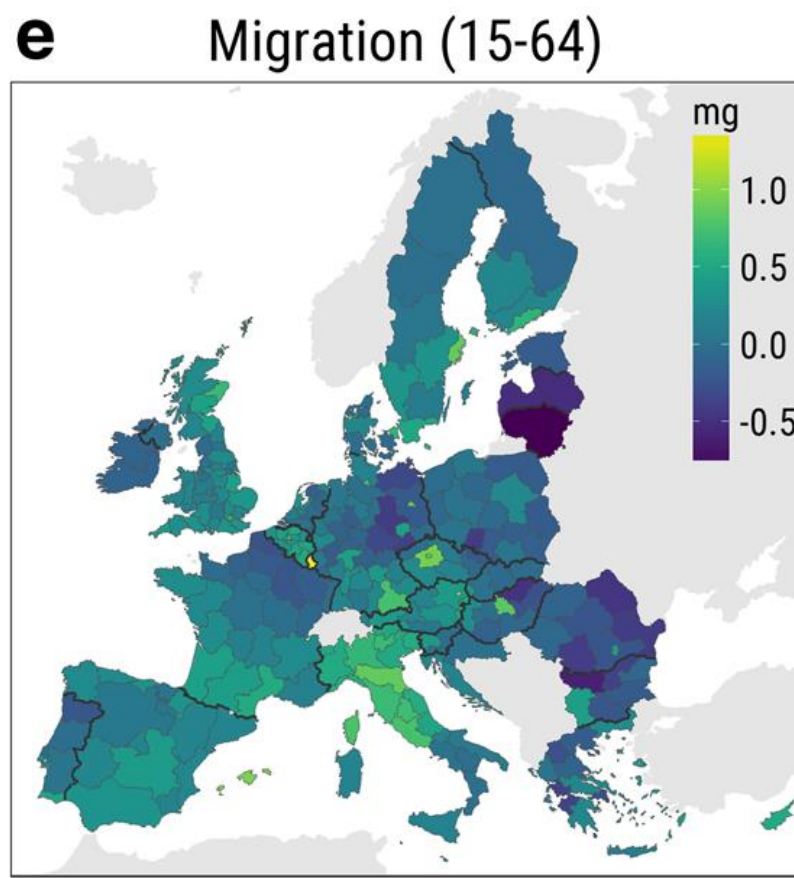
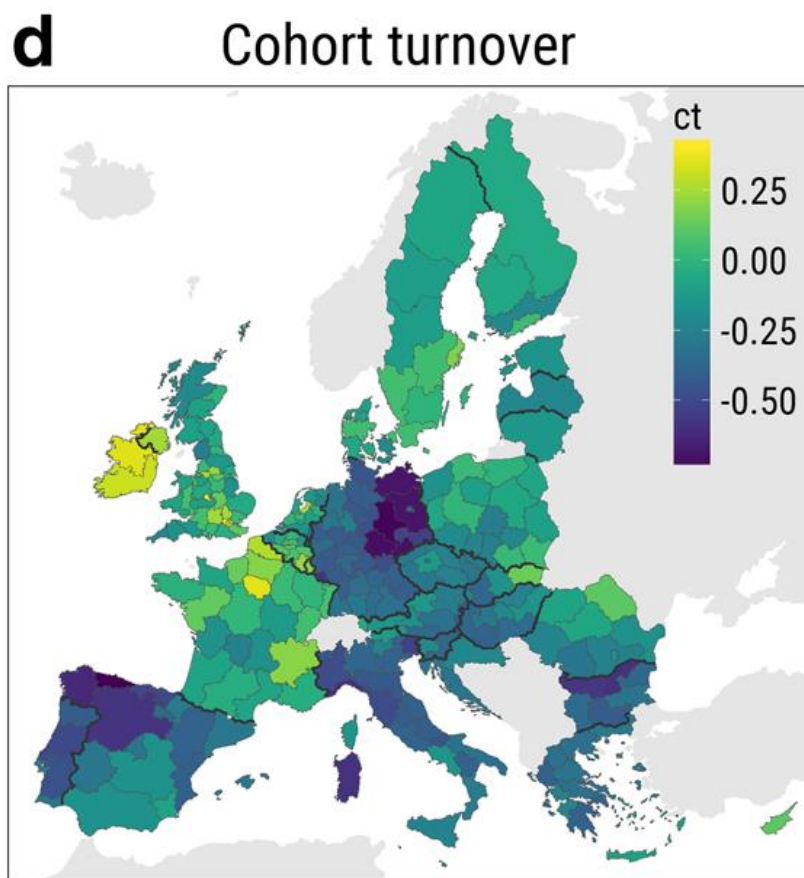


European population aging – Total Support Ratio



Kashnitsky et al. 2017

European population aging – Impact of demographic components on the rate of change of the working-age population



“The concept of age has become more complicated because life expectancy has increased and people at each age have had progressively more remaining years of life. As people modified their behavior to reflect these changes, 40-year olds began to act more like 30-year-olds had acted in the past.”

Sanderson & Scherbov (2007)



Sanderson & Scherbov approach: Years of life left

- **Chronological age** – number of years already past
- **Prospective age** – remaining years of life (remaining life expectancy)

A person can have 30 and 40 years at the same time

Year	Chronological age	Remaining Life expectancy (Prospective age)
1952	30	44.7
2005	30	54.4
2005	40	44.7

Source: Human Mortality Database used from Sanderson & Scherbov (2008)



Prospective Old-Age Dependency Ratio

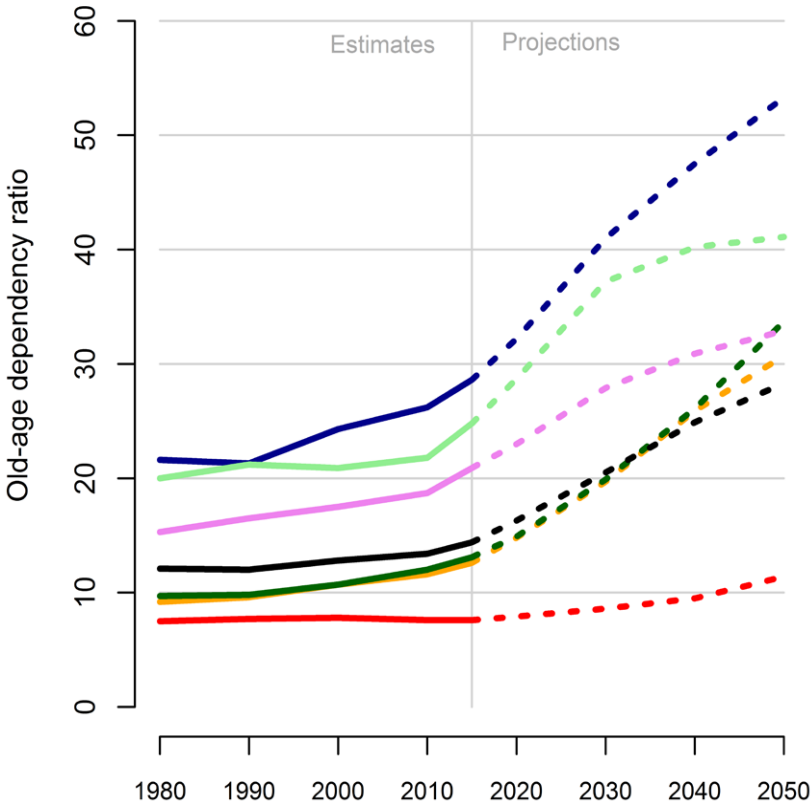
$$\text{POADR} = \frac{\text{Population older than the Old Age Threshold}}{\text{Population 15 – Old Age Threshold}} \times 100$$

The Old Age Threshold can be any age. Usually: 15 years left of Life expectancy

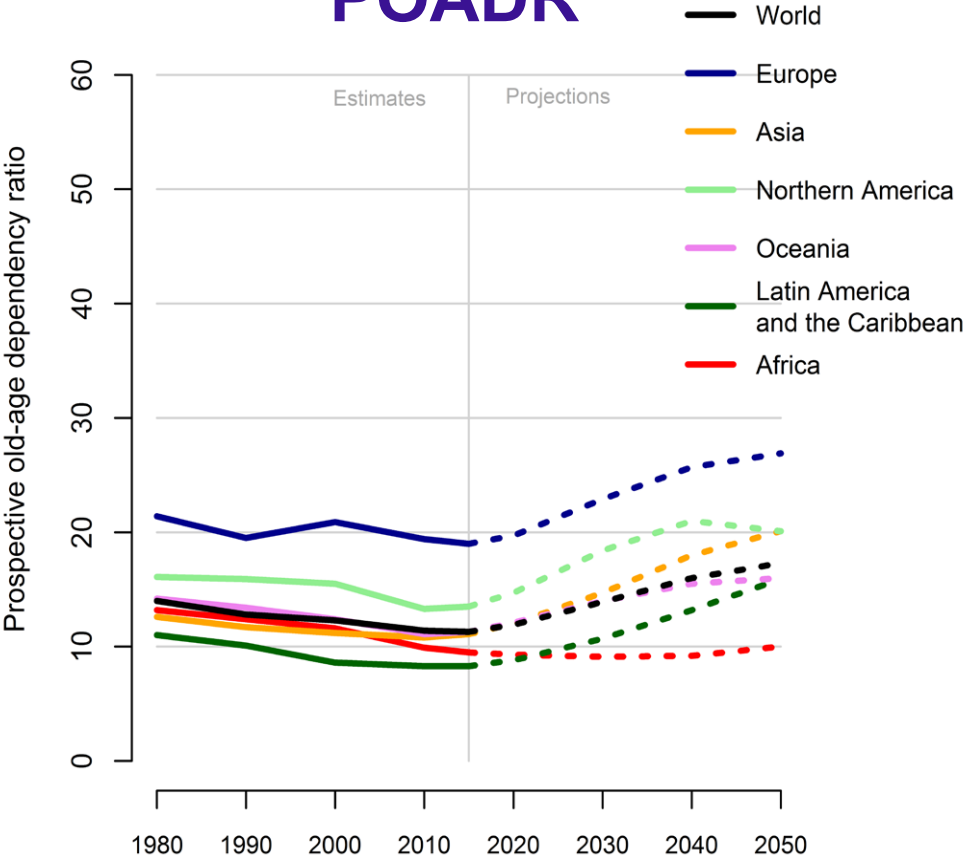


Prospective Old-Age Dependency Ratio

OADR



POADR



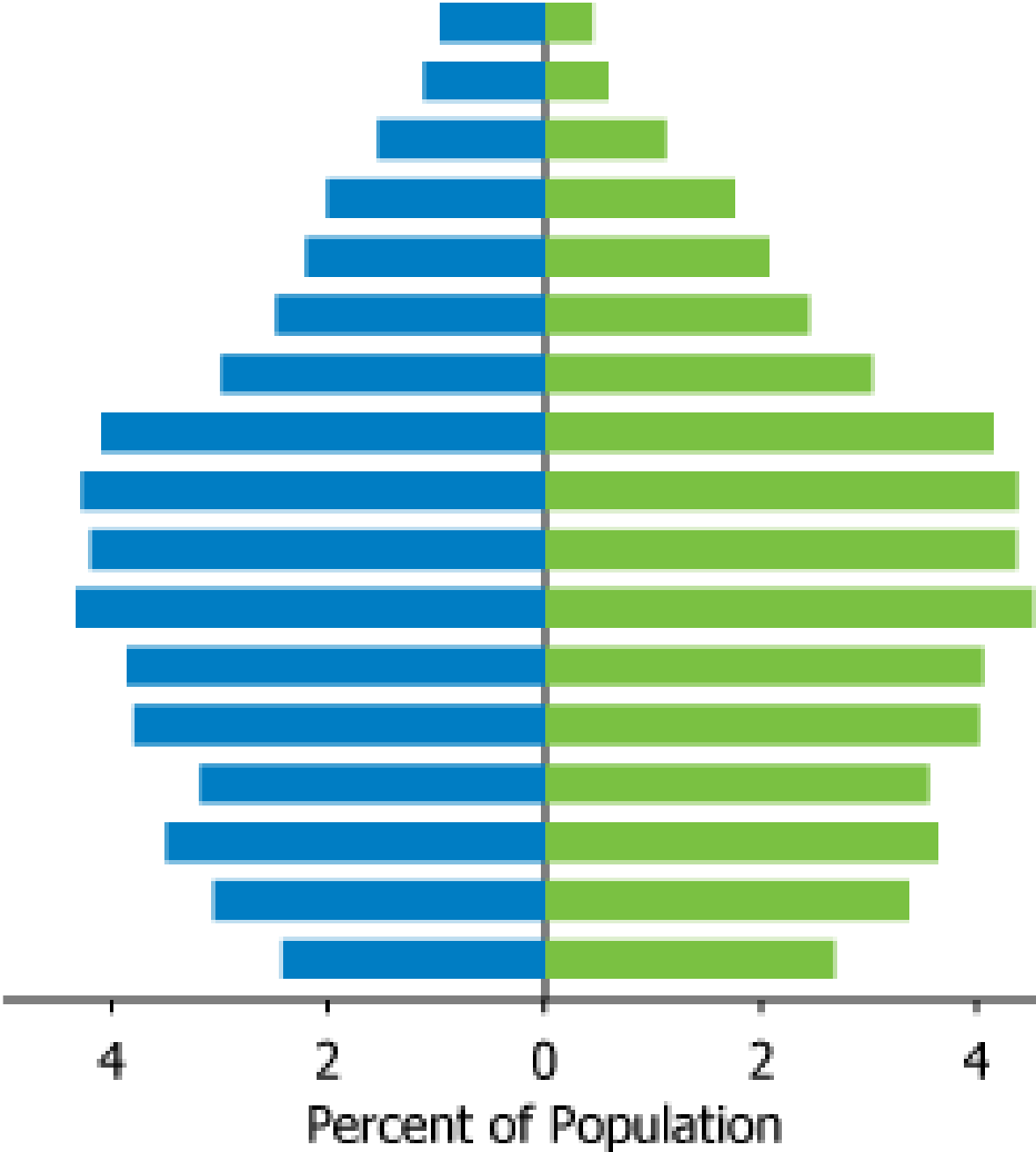
United Nations. World Population Aging, 2017

Economic approaches

1. Formal workers / Pensioners
2. Workers / Non-workers
3. Healthy / Non-healthy adults
4. Active / Non-active adults
5. Producers / Consumers



The Economic Support Ratio and the Demographic Dividend



Origin

- Asian economic growth in the 80`s and 90`s attached to low demographic dependency period → Bloom & Williamson (1998)
- Cutler et al. (1990) → Demographic window of opportunity
- While the young dependency ratio diminishes, frees more resources.
- Lower fertility rates are correlated with greater investments in human capital
- A usual measure to handle economic sustainability of population aging



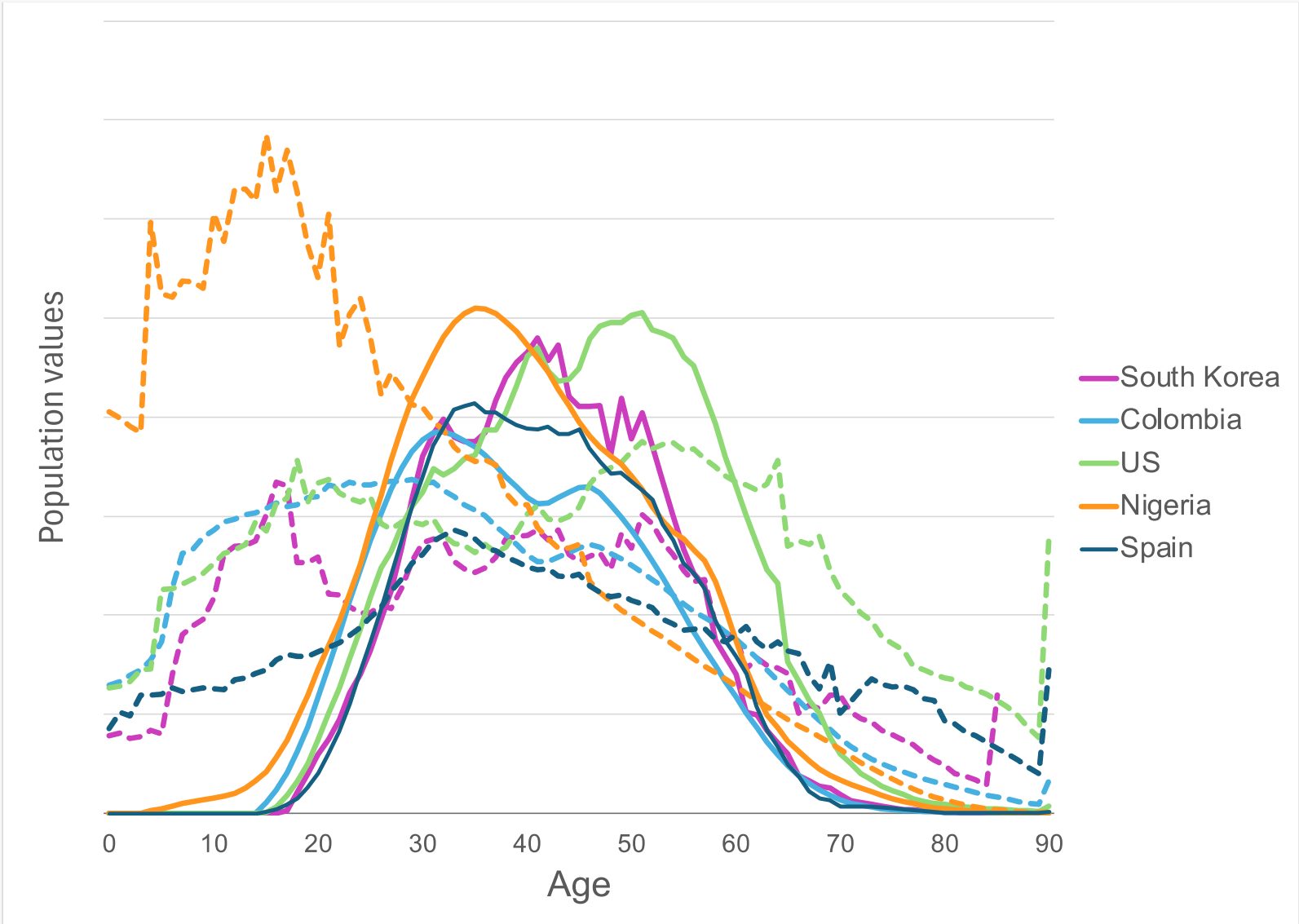
Economic Support Ratio

- Mason (2005) separates the population effect into two Demographic Dividends, and uses Consumption and Labor Income profiles to estimate an **Economic Support Ratio**

$$\underbrace{\frac{Y_t}{N_t}}_{\text{GDP per capita}} = \underbrace{\frac{L_t}{N_t}}_{\text{Economic Support Ratio}} \cdot \underbrace{\frac{Y_t}{L_t}}_{\text{Output per worker}}$$



Labor Income and Consumption profiles



Economic Support Ratio

- Mason (2005) separate the population effect into two Demographic Dividends, and use Consumption and Labor Income profiles to estimate an **Economic Support Ratio**

$$\underbrace{\frac{Y_t}{N_t}} = \underbrace{\frac{L_t}{N_t}} \cdot \underbrace{\frac{Y_t}{L_t}}$$

Effective Producers

Effective Consumers

Economic Support Ratio



Economic Support Ratio

Effective producers

$$L_t = \sum_a yl(a) \cdot P(a)$$

Effective consumers

$$N_t = \sum_a c(a) \cdot P(a)$$

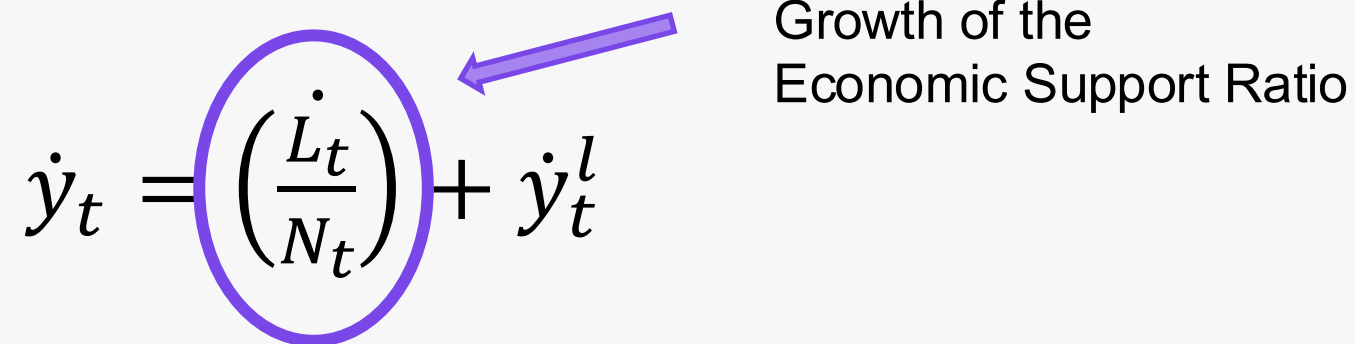


First Demographic dividend

The growth rate of the Economic Output per capita:

$$\dot{y}_t = \left(\frac{\dot{L}_t}{N_t} \right) + \dot{y}_t^l$$

Growth of the
Economic Support Ratio



While Producers grow at a faster rate than Consumers → There is a positive Demographic Dividend



Second Demographic dividend

The growth rate of the Economic Output per capita:

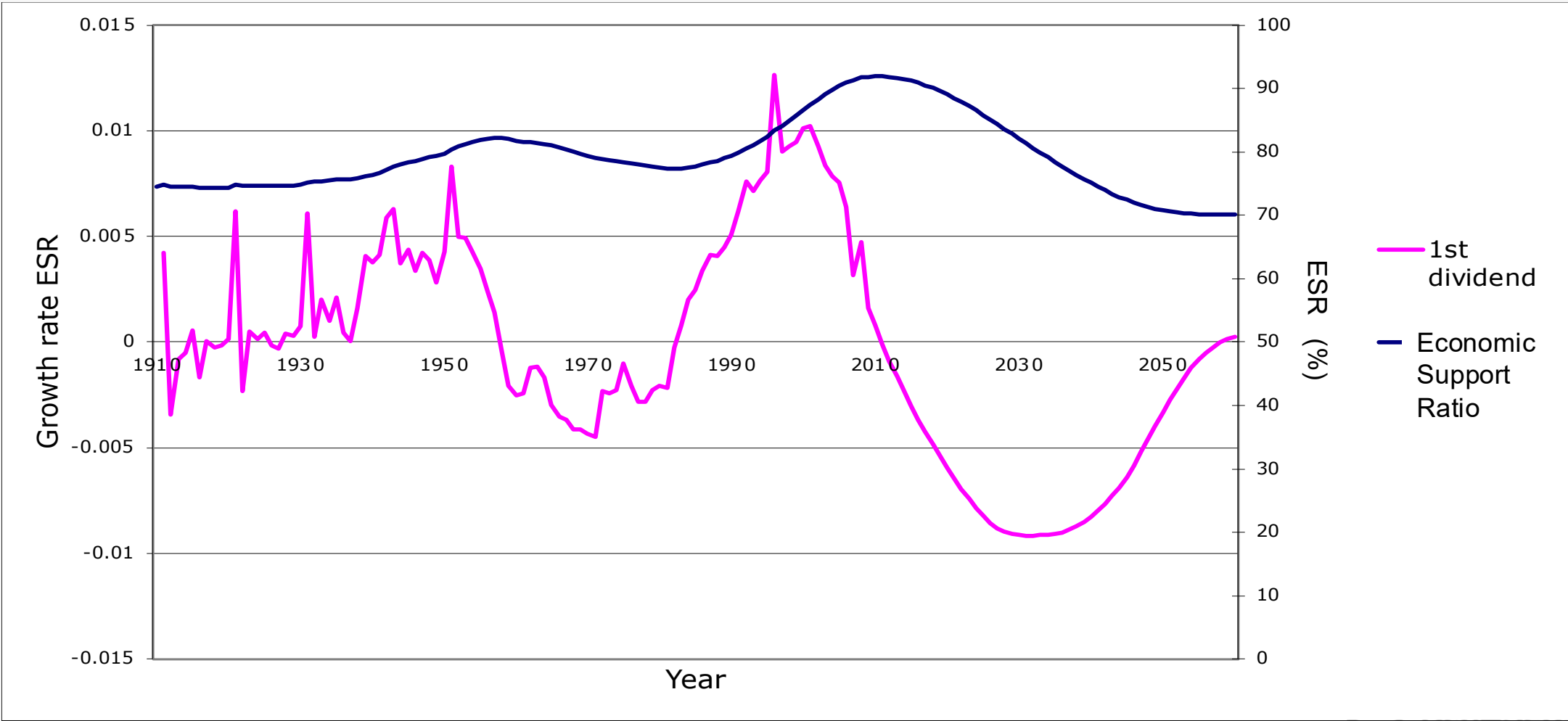
$$\dot{y}_t = \left(\frac{\dot{L}_t}{N_t} \right) + \dot{y}_t^l$$

Growth of the
Output per worker

- Related to Capital growth
- Fostered by Savings and Human Capital investments due to anticipated behavior



First Demographic dividend in Spain



Exercise



Estimating the demographic dividend

- Choose a country
- Data on population and economic profiles
- Obtain effective Producers and Consumers for each year

$$L_t = \sum_a yl(a) \cdot P(a)$$

$$N_t = \sum_a c(a) \cdot P(a)$$



Estimating the demographic dividend

- Estimate the Economic Support Ratio (ESR) for each year
- Estimate the growth rate of the ESR → demographic dividend

$$\left(\frac{\dot{L}_t}{N_t}\right) = \frac{L_t/N_t - L_{t-1}/N_{t-1}}{L_t/N_t} = \ln\left(\frac{L_t}{N_t}\right) - \ln\left(\frac{L_{t-1}}{N_{t-1}}\right)$$



Gràcies

Elisenda Rentería
erenteria@ced.uab.cat

Júlia Almeida Calazans
jalmeida@ced.uab.cat

Mariana Calderón-Jaramillo
mcalderon@ced.uab.cat

